

Newton Iteration.

Prop. Let $\alpha > 0$ be fixed. Given $x_1 > \sqrt{\alpha}$, define recursively

$$x_{n+1} = \frac{1}{2} \cdot \left(x_n + \frac{\alpha}{x_n} \right) \quad (n \in \mathbb{N})$$

Then, $\{x_n\}$ is monotonically decreasing and converges to $\sqrt{\alpha}$.

Proof. Observe that: for each $n \in \mathbb{N}$,

$$\frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) = x_{n+1} \leq x_n \iff \alpha \leq x_n^2 \quad \text{if } x_n > 0.$$

(Clearly, all $x_n > 0$, by definition, so using $\alpha \leq x_n^2$ for all $n \in \mathbb{N}$, by induction, For $n=1$, $\alpha \leq x_1^2$, since $\sqrt{\alpha} < x_1$. So, $x_2 \leq x_1$.

If $\alpha \leq x_n^2$, then

$$x_{n+1} = \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) \leq \frac{1}{2} \cdot (x_n + x_n) = x_n$$

and, $\frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) \geq \sqrt{x_n \cdot \frac{\alpha}{x_n}} = \sqrt{\alpha}$, so $\sqrt{\alpha} \leq x_{n+1} \leq x_n$.

So that, the equivalent condition with the induction gives that

$$x_{n+1} \leq x_n \quad \text{for all } n \in \mathbb{N}$$

That is, decrease monotonically, and $x_n \geq \sqrt{\alpha}$ for all $n \in \mathbb{N}$

Now, since $\{x_n\}$ is bounded below by $\sqrt{\alpha}$ and decreasing,

$\{x_n\}$ has a limit, denote $L = \lim_{n \rightarrow \infty} x_n$. Then,

$$L = \lim_{n \rightarrow \infty} x_{n+1} = \lim_{n \rightarrow \infty} \frac{1}{2} \left(x_n + \frac{\alpha}{x_n} \right) = \frac{1}{2} L + \frac{\alpha}{2L} \Rightarrow L = \sqrt{\alpha}.$$

So, proved $\square$

Note: Moreover, we can measure the 'error': put $\epsilon_n = |x_n - \sqrt{\alpha}| = x_n - \sqrt{\alpha}$.

Then, for each $n \in \mathbb{N}$, $\epsilon_{n+1} = \frac{\epsilon_n^2}{2x_n} < \frac{\epsilon_n^2}{2\sqrt{\alpha}}$ and that $\epsilon_{n+1} < 2\sqrt{\alpha} \cdot \left(\frac{\epsilon_1}{2\sqrt{\alpha}} \right)^{2^n}$.

Since $\epsilon_{n+1} = x_{n+1} - \sqrt{\alpha} = \frac{1}{2} \cdot \left(x_n + \frac{\alpha}{x_n} \right) - \sqrt{\alpha} = \frac{1}{2} \cdot \left(\epsilon_n + \sqrt{\alpha} + \frac{\alpha}{\epsilon_n + \sqrt{\alpha}} \right) - \sqrt{\alpha}$

$$= \frac{1}{2} \cdot \left(\frac{\epsilon_n^2}{\epsilon_n + \sqrt{\alpha}} \right) = \frac{\epsilon_n^2}{2x_n} < \frac{\epsilon_n^2}{2\sqrt{\alpha}}, \text{ so that}$$

$$\epsilon_n < \frac{1}{2\sqrt{\alpha}} \cdot \epsilon_{n-1}^2 < \frac{1}{2\sqrt{\alpha}} \cdot \left(\frac{1}{2\sqrt{\alpha}} \right)^{2^{n-1}} \cdot \epsilon_{n-2}^{2^2} \leq \dots \leq \left(\frac{1}{2\sqrt{\alpha}} \right)^{2^n - 1} \cdot \epsilon_1^{2^{n-1}} \quad \square$$